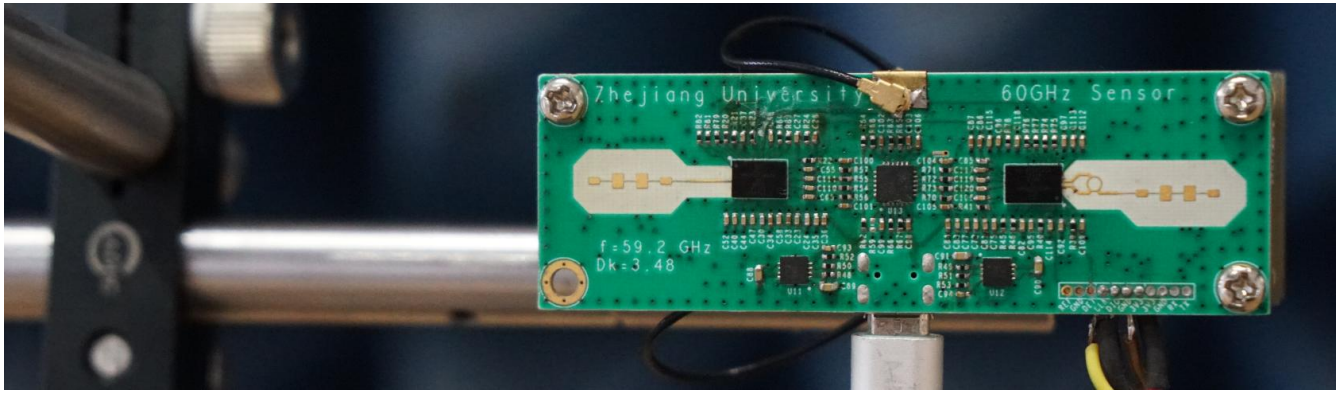
# Welcome to RF world




Specialized course

## Radio frequency circuits and systems, lecture 13

6:30~8:30 pm, Friday

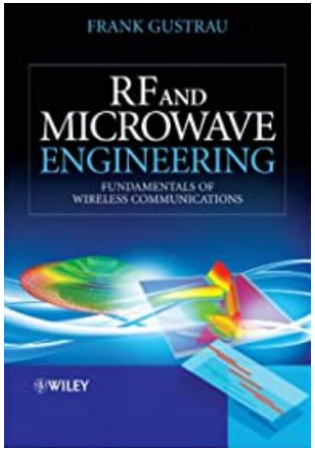
Room 102, No. 7 Building, Yuquan Campus, ZJU

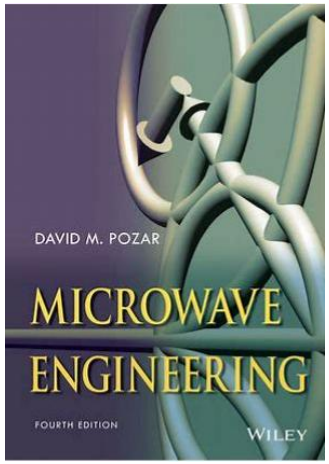
Lixin Ran

Center for advanced radiofrequency engineering (ARE), ISEE

https://person.zju.edu.cn/ranlx

# Text books




**RF AND MICROWAVE ENGINEERING**

**FUNDAMENTALS OF WIRELESS COMMUNICATIONS**

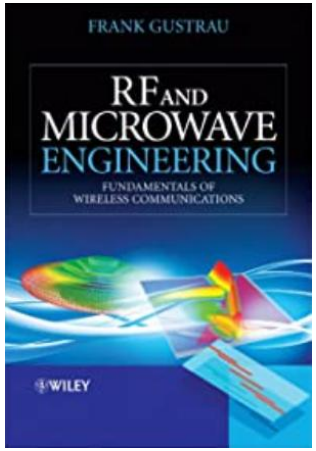
**Frank Gustrau**

*Dortmund University of Applied Sciences and Arts, Germany*

WILEY

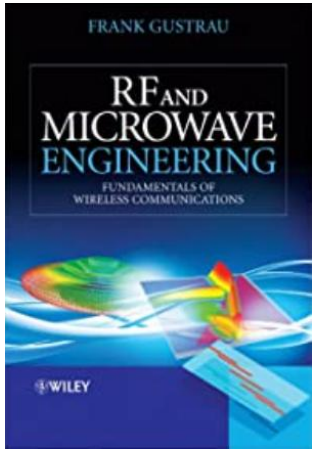
A John Wiley & Sons, Ltd., Publication

# Chapter 1




**1 Introduction**
1.1 Radiofrequency and Microwave Applications
1.2 Frequency Bands
1.3 Physical Phenomena in the High Frequency Domain
*1.3.1 Electrically Short Transmission Line*
*1.3.2 Transmission Line with Length Greater than One-Tenth of Wavelength*
*1.3.3 Radiation and Antennas*
1.4 Outline of the Following Chapters
References

# Chapter 2





**2 Electromagnetic Fields and Waves**

2.1 Electric and Magnetic Fields
- *2.1.1 Electrostatic Fields*
- *2.1.2 Steady Electric Current and Magnetic Fields*
- *2.1.3 Differential Vector Operations*

2.2 Maxwell's Equations
- *2.2.1 Differential Form in the Time Domain*
- *2.2.2 Differential Form for Harmonic Time Dependence*
- *2.2.3 Integral Form*
- *2.2.4 Constitutive Relations and Material Properties*
- *2.2.5 Interface Conditions*

2.3 Classification of Electromagnetic Problems
- *2.3.1 Static Fields*
- *2.3.2 Quasi-Static Fields*
- *2.3.3 Coupled Electromagnetic Fields*

2.4 Skin Effect

2.5 Electromagnetic Waves
- *2.5.1 Wave Equation and Plane Waves*
- *2.5.2 Polarization of Waves*
- *2.5.3 Reflection and Refraction*
- *2.5.4 Spherical Waves*

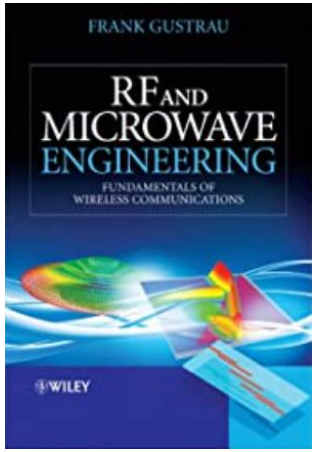
2.6 Summary

2.7 Problems

References

Further Reading

# Chapter 3





**3 Transmission Line Theory and Transient Signals on Lines**

3.1 Transmission Line Theory

*3.1.1 Equivalent Circuit of a Line Segment*

*3.1.2 Telegrapher's Equation*

*3.1.3 Voltage and Current Waves on Transmission Lines*

*3.1.4 Load-Terminated Transmission Line*

*3.1.5 Input Impedance*

*3.1.6 Loss-less Transmission Lines*

*3.1.7 Low-loss Transmission Lines*

*3.1.8 Transmission Line with Different Terminations*

*3.1.9 Impedance Transformation with Loss-less Lines*

*3.1.10 Reflection Coefficient*

*3.1.11 Smith Chart*

3.2 Transient Signals on Transmission Lines

*3.2.1 Step Function*

*3.2.2 Rectangular Function*

3.3 Eye Diagram

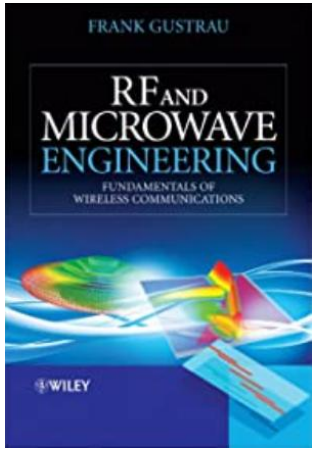
3.4 Summary

3.5 Problems

References

Further Reading

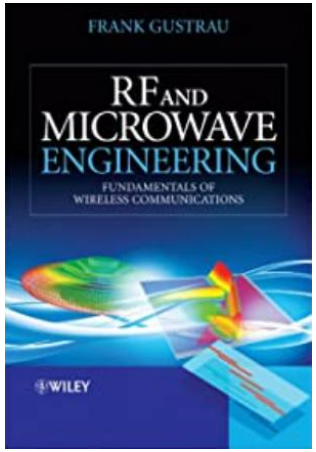
# Chapter 4





**4 Transmission Lines and Waveguides**
4.1 Overview
4.2 Coaxial Line
*4.2.1 Specific Inductance and Characteristic Impedance*
*4.2.2 Attenuation of Low-loss Transmission Lines*
*4.2.3 Technical Frequency Range*
*4.2.4 Areas of Application*
4.3 Microstrip Line
*4.3.1 Characteristic Impedance and Effective Permittivity*
*4.3.2 Dispersion and Technical Frequency Range*
*4.3.3 Areas of Application*
4.4 Stripline
*4.4.1 Characteristic Impedance*
*4.4.2 Technical Frequency Range*
4.5 Coplanar Line
*4.5.1 Characteristic Impedance and Effective Permittivity*
*4.5.2 Coplanar Waveguide over Ground*
*4.5.3 Coplanar Waveguides and Air Bridges*
*4.5.4 Technical Frequency Range*
*4.5.5 Areas of Application*
4.6 Rectangular Waveguide
*4.6.1 Electromagnetic Waves between Electric Side Walls*
*4.6.2 Dominant Mode (TE10)*
*4.6.3 Higher Order Modes*
*4.6.4 Areas of Application*
*4.6.5 Excitation of Waveguide Modes*
*4.6.6 Cavity Resonators*
4.7 Circular Waveguide
4.8 Two-Wire Line
*4.8.1 Characteristic Impedance*
*4.8.2 Areas of Application*
4.9 Three-Conductor Transmission Line
*4.9.1 Even and Odd Modes*
*4.9.2 Characteristic Impedances and Propagation Constants*
*4.9.3 Line Termination for Even and Odd Modes*
4.10 Problems
References

# Chapter 5





**5 Scattering Parameters**

5.1 Multi-Port Network Representations

5.2 Normalized Power Waves

5.3 Scattering Parameters and Power

5.4 S-Parameter Representation of Network Properties

*5.4.1 Matching*

*5.4.2 Complex Conjugate Matching*

*5.4.3 Reciprocity*

*5.4.4 Symmetry*

*5.4.5 Passive and Loss-less Circuits*

*5.4.6 Unilateral Circuits*

*5.4.7 Specific Characteristics of Three-Port Networks*

5.5 Calculation of S-Parameters

*5.5.1 Reflection Coefficients*

*5.5.2 Transmission Coefficients*

*5.5.3 Renormalization*

5.6 Signal Flow Method

*5.6.1 One-Port Network/Load Termination*

*5.6.2 Source*

*5.6.3 Two-Port Network*

*5.6.4 Three-Port Network*

*5.6.5 Four-Port Network*

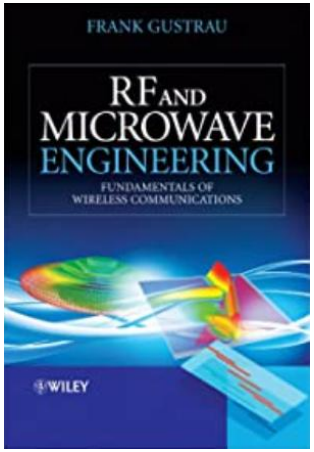
5.7 S-Parameter Measurement

5.8 Problems

References

Further Reading

# Chapter 6





**6 RF Components and Circuits**

6.1 Equivalent Circuits of Concentrated Passive Components
- *6.1.1 Resistor*
- *6.1.2 Capacitor*
- *6.1.3 Inductor*

6.2 Transmission Line Resonator
- *6.2.1 Half-Wave Resonator*
- *6.2.2 Quarter-Wave Resonator*

6.3 Impedance Matching
- *6.3.1 LC-Networks*
- *6.3.2 Matching Using Distributed Elements*

6.4 Filter
- *6.4.1 Classical LC-Filter Design*
- *6.4.2 Butterworth Filter*

6.5 Transmission Line Filter
- *6.5.1 Edge-Coupled Line Filter*
- *6.5.2 Hairpin Filter*
- *6.5.3 Stepped Impedance Filter*
- *6.5.4 Parasitic Box Resonance*
- *6.5.5 Waveguide Filter*

6.6 Circulator

6.7 Power Divider
- *6.7.1 Wilkinson Power Divider*
- *6.7.2 Unequal Split Power Divider*

6.8 Branchline Coupler
- *6.8.1 Conventional 3 dB Coupler*
- *6.8.2 Unequal Split Branchline Coupler*

6.9 Rat Race Coupler

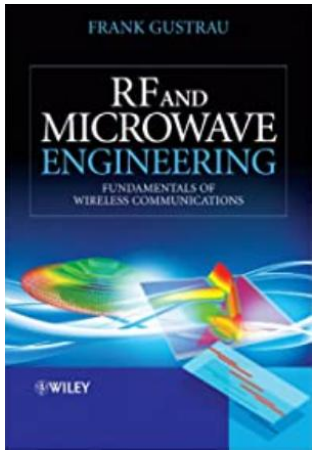
6.10 Directional Coupler

6.11 Balanced-to-Unbalanced Circuits

6.12 Electronic Circuits
- *6.12.1 Mixers*
- *6.12.2 Amplifiers and Oscillators*

6.13 RF Design Software
- *6.13.1 RF Circuit Simulators*
- *6.13.2 Three-Dimensional Electromagnetic Simulators*

# Chapter 7





**7 Antennas**

7.1 Fundamental Parameters
- *7.1.1 Nearfield and Farfield*
- *7.1.2 Isotropic Radiator*
- *7.1.3 Radiation Pattern and Related Parameters*
- *7.1.4 Impedance Matching and Bandwidth*

7.2 Standard Types of Antennas

7.3 Mathematical Treatment of the Hertzian Dipole

7.4 Wire Antennas
- *7.4.1 Half-Wave Dipole*
- *7.4.2 Monopole*
- *7.4.3 Concepts for Reducing Antenna Height*

7.5 Planar Antennas
- *7.5.1 Rectangular Patch Antenna*
- *7.5.2 Circularly Polarizing Patch Antennas*
- *7.5.3 Planar Dipole and Inverted-F Antenna*

7.6 Antenna Arrays
- *7.6.1 Single Element Radiation Pattern and Array Factor*
- *7.6.2 Phased Array Antennas*
- *7.6.3 Beam Forming*

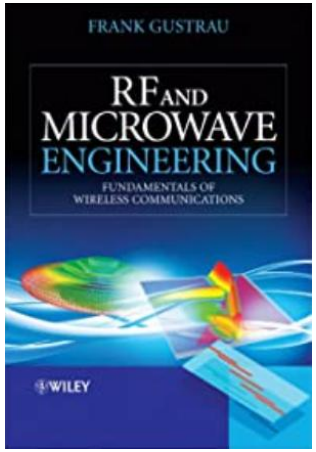
7.7 Modern Antenna Concepts

7.8 Problems

References

Further Reading

# Chapter 8





**8 Radio Wave Propagation**

8.1 Propagation Mechanisms

*8.1.1 Reflection and Refraction*

*8.1.2 Absorption*

*8.1.3 Diffraction*

*8.1.4 Scattering*

*8.1.5 Doppler Effect*

8.2 Basic Propagation Models

*8.2.1 Free Space Loss*

*8.2.2 Attenuation of Air*

*8.2.3 Plane Earth Loss*

*8.2.4 Point-to-Point Radio Links*

*8.2.5 Layered Media*

8.3 Path Loss Models

*8.3.1 Multipath Environment*

*8.3.2 Clutter Factor Model*

*8.3.3 Okumura–Hata Model*

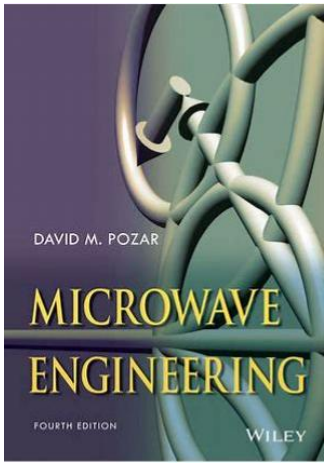
*8.3.4 Physical Models and Numerical Methods*

8.4 Problems

References

Further Reading

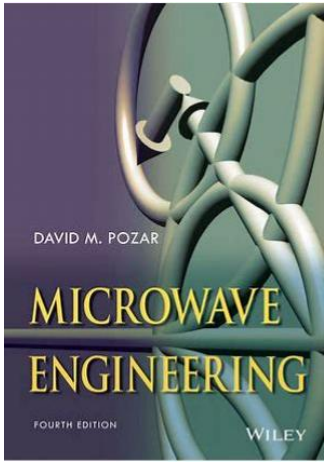
# Introduction: Text book




C h a p t e r N i n e

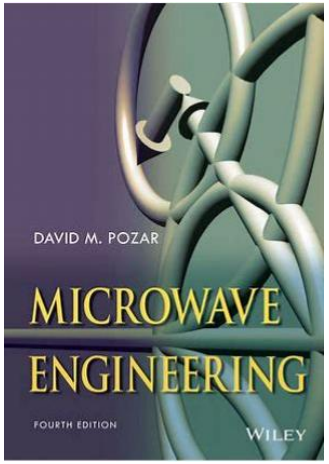
Theory and Design
of Ferrimagnetic Components

# Introduction: Text book



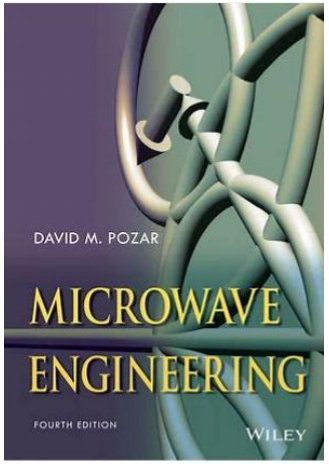

Chapter 10: NOISE AND NONLINEAR DISTORTION
- 10.1 Noise in Microwave Circuits
  - Dynamic Range and Sources of Noise
  - Noise Power and Equivalent Noise Temperature
  - Measurement of Noise Temperature
- 10.2 Noise Figure
  - Definition of Noise Figure
  - Noise Figure of a Cascaded System
  - Noise Figure of a Passive Two-Port Network
  - Noise Figure of a Mismatched Lossy Line
  - Noise Figure of a Mismatched Amplifier
- 10.3 Nonlinear Distortion
  - Gain Compression
  - Harmonic and Intermodulation Distortion
  - Third-Order Intercept Point
  - Intercept Point of a Cascaded System
  - Passive Intermodulation
- 10.4 Dynamic Range
  - Linear and Spurious Free Dynamic Range

# Introduction: Text book



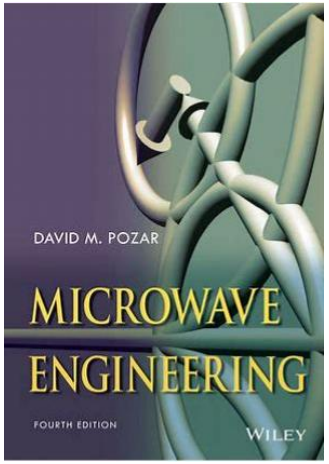

Chapter 11: ACTIVE RF AND MICROWAVE DEVICES
- 11.1 Diodes and Diode Circuits
  - Schottky Diodes and Detectors
  - PIN Diodes and Control Circuits
  - Varactor Diodes
  - Other Diodes
  - Power Combining
- 11.2 Bipolar Junction Transistors
  - Bipolar Junction Transistor
  - Heterojunction Bipolar Transistor
- 11.3 Field Effect Transistors
  - Metal Semiconductor Field Effect Transistor
  - Metal Oxide Semiconductor Field Effect Transistor
  - High Electron Mobility Transistor
- 11.4 Microwave Integrated Circuits
  - Hybrid Microwave Integrated Circuits
  - Monolithic Microwave Integrated Circuits
- 11.5 Microwave Tubes

# Introduction: Text book




Chapter 12: MICROWAVE AMPLIFIER DESIGN
- 12.1 Two-Port Power Gains
  - Definitions of Two-Port Power Gains
  - Further Discussion of Two-Port Power Gains
- 12.2 Stability
  - Stability Circles
  - Tests for Unconditional Stability
- 12.3 Single-Stage Transistor Amplifier Design
  - Design for Maximum Gain (Conjugate Matching)
  - Constant-Gain Circles and Design for Specified Gain
  - Low-Noise Amplifier Design
  - Low-Noise MOSFET Amplifier
- 12.4 Broadband Transistor Amplifier Design
  - Balanced Amplifiers
  - Distributed Amplifiers
  - Differential Amplifiers
- 12.5 Power Amplifiers
  - Characteristics of Power Amplifiers and Amplifier Classes
  - Large-Signal Characterization of Transistors
  - Design of Class A Power Amplifiers

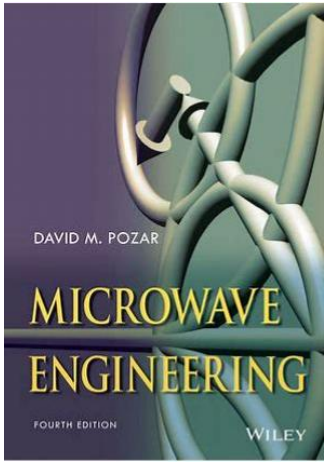
# Introduction: Text book





- Chapter 13: OSCILLATORS AND MIXERS
  - 13.1 RF Oscillators
    - General Analysis
    - Oscillators Using a Common Emitter BJT
    - Oscillators Using a Common Gate FET
    - Practical Considerations
    - Crystal Oscillators
  - 13.2 Microwave Oscillators
    - Transistor Oscillators
    - Dielectric Resonator Oscillators
  - 13.3 Oscillator Phase Noise
    - Representation of Phase Noise
    - Leeson's Model for Oscillator Phase Noise
  - 13.4 Frequency Multipliers
    - Reactive Diode Multipliers (Manley–Rowe Relations)
    - Resistive Diode Multipliers
    - Transistor Multipliers
  - 13.5 Mixers
    - Mixer Characteristics
    - Single-Ended Diode Mixer
    - Single-Ended FET Mixer
    - Balanced Mixer
    - Image Reject Mixer
    - Differential FET Mixer and Gilbert Cell Mixer
    - Other Mixers

# Chapter 14





- 14.1 System Aspects of Antennas
  - Fields and Power Radiated by an Antenna
  - Antenna Pattern Characteristics
  - Antenna Gain and Efficiency
  - Aperture Efficiency and Effective Area
  - Background and Brightness Temperature
  - Antenna Noise Temperature and G/T
- 14.2 Wireless Communications
  - The Friis Formula
  - Link Budget and Link Margin
  - Radio Receiver Architectures
  - Noise Characterization of a Receiver
  - Digital Modulation and Bit Error Rate
  - Wireless Communication Systems
- 14.3 Radar Systems
  - The Radar Equation
  - Pulse Radar
  - Doppler Radar
  - Radar Cross Section
- 14.4 Radiometer Systems
  - Theory and Applications of Radiometry
  - Total Power Radiometer
  - The Dicke Radiometer
- 14.5 Microwave Propagation
  - Atmospheric Effects
  - Ground Effects
  - Plasma Effects
- 14.6 Other Applications and Topics
  - Microwave Heating
  - Power Transfer
  - Biological Effects and Safety

# Examination Paper

- **Concepts:**
  - Fill in blanks
  - 35 blanks
  - 70 points in total
- **Calculations:**
  - 3 examination questions
  - 30 points in total
- **Time:**
  - 120 minutes

# Reminder

- **Examination**

## 浙江大学 2025－2026 学年秋冬学期
## 《射频电路与系统》课程期末考试试卷

课程号：__67190130_，开课学院：__信息与电子工程学院__

考试试卷：A 卷 √、B 卷（请在选定项上打 √）

考试形式：闭、开卷 √（请在选定项上打 √），允许带本课程的教材、个人笔记、字典、计算器、圆规、尺子等入场

考试日期：_2026_年_01_月_13_日，考试时间：_120_分钟

- **When/Where　2026-1-13 8:00---10:00　教4-312**

# Reminder

- 开卷考试：教材、作业（1.10截止）、计算器

  可选：笔记、字典、课程PPT
- 试题内容基于教材
- 重要：考前一定要过一遍教材，熟悉教材脉络

  考试时仔细读题，查找关键词
- 填空题原则上英文答题，大题建议英文答题，不要空题！！！
- 考试时间是足够的，遇到找不到的题目先跳过，最大化得分
- 考试结束将回收租借的教材，想留下教材也可以现场购买

# Acknowledgement

**Thank you for listening!**

**祝大家考试顺利！**